SwarupAnand's PHYSICS

The Falling Apple Everyday, Keeps Confusion Away !!!

Batch : 11th - (2025-2027)

Time : 1 hr.

**Date : 06/01/2026** **Chapters:COM & Circular Motion** **Marks: 100**

1. A particle moves along a circle of radius R with a constant angular speed $\omega$. Its displacement (only magnitude ) in time t will be
(a) $\omega t$ (b) $2R \cos \omega t$
(c) $2R \sin \omega t$ (d) $2R \sin \omega t/2$

2. Two racing cars of masses $m_1$ and $m_2$ are moving in circles of radii $r_1$ and $r_2$ respectively. Their speeds are such that each makes a complete circle in the same duration of time t. The ratio of the angular speeds of the first to the second car is
(a) $m_1 : m_2$ (b) $r_1 : r_2$
(c) 1 ; 1 (d) $m_1r_1 : m_2r_2$

3. The ratio of angular speeds of minute hand and hour hand of a watch is
(a) 1 : 12 (b) 6 : 1 (c) 12 : 1 (d) 1 : 6

4. The angular speed of seconds needle in a mechanical watch is
(a) $\frac{\pi}{30} rad/s$ (b) $2\pi rad/s$
(c) $\pi rad/s$ (d) $\frac{60}{\pi} rad/s$

5. If $a_r$ and $a_t$ represent radial and tangential accelerations, the motion of a particle will be uniformly circular if
(a) $a_r = 0$ and $a_t = 0$ (b) $a_r = 0$ but $a_r \neq 0$
(c) $a_r \neq 0$ and $a_t = 0$ (d) $a_r \neq 0$ and $a_t \neq 0$

6. A body is whirled in a horizontal circle of radius 20 cm. It has an angular velocity of 10 rad/s. what is its linear velocity at any point on circular path?
(a) 20 m/s (b) $\sqrt{2} m/s$
(c) 10 m/s (d) 2 m/s

7. An electric fan has blades of length 30 cm measured from the axis of rotation. If the fan is rotating at 120 rpm, the acceleration of a point on the tip of the blade is
(a) 1600 $ms^{-2}$ (b) 47.4 $ms^{-2}$
(c) 23.7 $ms^{-2}$ (d) 50.55 $ms^{-2}$

8. A partical moves in a circle of radius 25 cm at two revolutions per second. The acceleration of the particle in $m/s^2$ is
(a) $\pi^2$ (b) $8\pi^2$ (c) $4\pi^2$ (d) $2\pi^2$

9. If the equation for the displacement of a particle moving on a circular path is given by $(\theta) = 2t^3 + 0.5$, where $\theta$ us ub radians and t in seconds, then the angular velocity of the particle after 2s its start is
(a) 8 rad/s (b)12 rad/s (c)24rad/s (d)36rad/s

10. If angular velocity of a disc depends an angle rotated $\theta$ as $\omega = \theta^2 + 2\theta$, then its angular acceleration $\alpha$ at $\theta = 1$ rad is
(a) 8 rad/$sec^2$ (b) 10 rad/$sec^2$
(c) 12 rad/$sec^2$ (d) none

11. A string breaks if its tension exceeds 10 newtons, A stone of mass 250 gm tied to this string of length 10 cm is rotated in a horizontal circle. The maximum angular velocity of rotation can be
(a) 20 rad/s (b) 40 rad/s
(c) 100 rad/s (d) 200 rad/s

12. A bob of mass 10 kg is attached to wire 0.3 m long. Its breaking stress is $4.8 \times 10^7$ N/$m^2$. The area of cross section of the wire is $10^{-6}$ $m^2$. The maximum angular velocity with which it can be rotated in a horizontal circle
(a) 8 rad/sec (b) 4 rad/sec
(c) 2 rad/sec (d) 1 rad/sec

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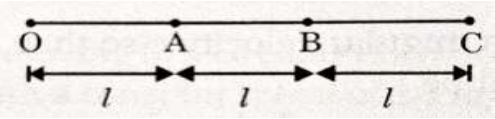
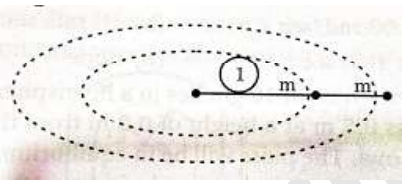
13. Three identical particles are joined together by a thread as shown in figure. All the three particles are moving in a horizontal plane. If the velocity of the outermost particle is $\upsilon_0$, then the ratio of tensions in the three sections of the string is : ($T_{BC} : T_{AB} : T_{OA}$)



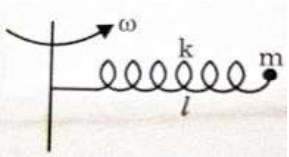

(a) 3 : 5 : 7 (b) 3 : 4 : 5
(c) 7 : 11 : 6 (d) 3 : 5 : 6

14. Two particles each of mass m are moving in horizontal circle with same angular speed. If both string are of same length then the ratio of tension in string $\frac{r_1}{r_2}$ is




(a) 3/2 (b) 3 (c) 2 (d) 1/3

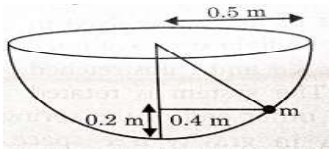
15. A particle of mass m is fixed to one end of a light spring of force constsant k and unstretched length $l$. The system is rotated about the other end of the spring with an angular velocity $\omega$, in gravity free space. The increase in length of the spring will be




(a) $\frac{m\omega^2 l}{k}$ (b) $\frac{m\omega^2 l}{k - m\omega^2}$

(c) $\frac{m\omega^2 l}{k + m\omega^2}$ (d) none

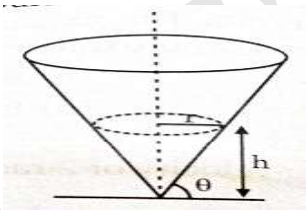
16. A small mass of 10 gm lies in a hemispherical bowl of radius 0.5 at a height of 0.2 m from the bottom of the bowl. The mass will be in equillbrium of the bowl rotates at an angular speed of : ( g = 10 m/sec$^2$)




(a) $\frac{10}{\sqrt{3}} rad/s$ (b) $10\sqrt{3} rad/s$

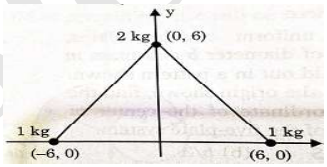
(c) 10 rad/s (d) $\sqrt{20} rad/s$

17. A particle describes a horizontal circle of radius r on the smooth surface of an inverted cone as shown . The height of plane of circle above vertex is h. The speed of particle should be




(a) $\sqrt{rg}$ (b) $\sqrt{2rg}$ (c) $\sqrt{gh}$ (d) $\sqrt{2gh}$

18. Centre of mas of system will be




(a) (0, 4) (b) (0, 3) (c) ( 0, 2 ) (d) (3, 0)

19. Two bodies of mass 1 kg and 3 kg have position vectors $\hat{i} + 2\hat{j} + \hat{k} and -3\hat{i} - 2\hat{j} + \hat{k}$, respectively. The center of mass of this system has a position vector

(a) $-2\hat{i} + 2\hat{k}$ (b) $-2\hat{i} - \hat{j} + \hat{k}$

(c) $-2\hat{i} - \hat{j} - 2\hat{k}$ (d) $-\hat{i} - \hat{j} - \hat{k}$

20. Four particles of masses m, 2m, 4m and 8m are placed at the corners of a square of side a. The co-ordinates of centre of mass is

(a) $[\frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}]$ (b) $[\frac{a}{2}, \frac{3a}{4}]$

(c) $[\frac{a}{5}, \frac{2a}{5}]$ (d) $[\frac{2a}{5}, \frac{4a}{5}]$

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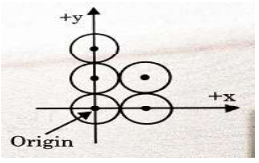
21. Five uniform circular plates, each of diameter b and mass m are laid out in a pattern shown. Using the origin shown , find the y co-ordinate of the center of mass of the five-plate system




(a) b/5
(b) b/3
(c) 4b/5
(d) 2b/5

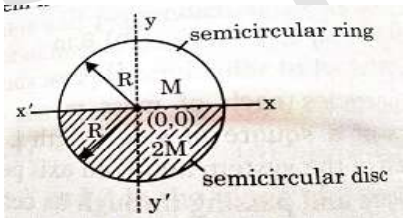
22. If linear density of a rod of length 3m varies as $\lambda = 2 + x$, then the position of the centre of gravity of the rod is

(a) $\frac{7}{3}m$ (b) $\frac{12}{7}m$ (c) $\frac{10}{7}m$ (d) $\frac{9}{7}m$

23. The mass per unit length of a non- uniform rod of length L is given by $\mu = \lambda x^2$, where $\lambda$ is a constant and x is distance from one end of the rod. The distance of the center of mass of rod from this end is

(a) $\frac{L}{2}$ (b) $\frac{L}{4}$ (c) $\frac{3L}{4}$ (d) $\frac{L}{3}$

24. A system is formed as shown below

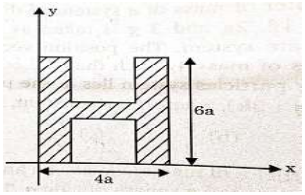



The centre of mass of the system is

(a) $[\frac{2R}{8\pi}, 0]$ (b) $[0, -\frac{2R}{9\pi}]$

(c) $[0, \frac{9\pi}{2R}]$ (d) $[\frac{8\pi}{2R}, 0]$

25. An object of mass M having shape like 'H' of width 4a & height 6a is placed as shown then coordinate of centre of mass will be




(a) (2a, 2a) (b) (a, 3a)
(c) (2a, 3a) (d) (3a, 2a)

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